

Local Lipschitz continuity of the canonical Taylor-tree coefficients

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Abstract

Astra #11 rank 1. Composing the amplitude Lipschitz estimate (unit 143) with the bounded-linear difference form of the canonical coefficients (units 140–141) gives, on the bounded family $\|x\|_\rho, \|x'\|_\rho \leq M_x, \|y\|_\rho, \|y'\|_\rho \leq M_y$,

$$|A_\alpha(x, y) - A_\alpha(x', y')| \leq \text{lipA} \cdot (\|y - y'\|_\rho + 2\beta M_y \|x - x'\|_\rho),$$

and the same bound for $|B_\alpha(x, y) - B_\alpha(x', y')|$ with the constant lipB , with explicit constants (`abs_canonA_ampCoeff_sub_le`, `abs_canonB_ampCoeff_sub_le`). This is the chart-level formal counterpart of the continuity of $\xi \mapsto C_{\mu, m}(\xi)$ that §4.3 of the paper asserts by reference to the continuous mapping theorem. Zero `sorry/axiom`.

1 The things formalised

```
theorem tsum_geom_prod ( ) (h0 : 0 < ) (h1 : < 1) :
  ' ij : x , ^ ij.1 * ^ ij.2 = (1 - ) ^ 2
theorem cWeight_nonneg / uWeight_nonneg / vWeight_nonneg
theorem ampEnv_le_common ... : ampEnv x y s My * (1 + s) ^ 0 * exp ( * s * (2 * Mx))
theorem ampCoeff_sub_env ... (hs : 0 < s) :
  |ampCoeff x y ij s - ampCoeff x' y' ij s|
  ( ^ (ij.1 + ij.2)) ^ 1 * ((Dy + 2 * * My * Dx) * gaussEnv (2 * Mx) 1 s)
theorem tailMoment_ampCoeff_sub_le ... :
  tailMoment (fun s => ampCoeff x y ij s - ampCoeff x' y' ij s)
  ( ^ (ij.1 + ij.2)) ^ 1 * (Dy + 2 * * My * Dx)
  * envMoment (gaussEnv (2 * Mx) 1)
theorem coeffNorm_ampCoeff_sub_le ... (hr0 : 0 < r) (hr : r < ) :
  coeffNorm r (fun ij s => ampCoeff x y ij s - ampCoeff x' y' ij s)
  (1 - r / ) ^ 1 ^ 2 * (Dy + 2 * * My * Dx) * envMoment (gaussEnv (2 * Mx) 1)
def lipA ( r) (h h k k) (Mx ) := cWeight r ... * ((1 - r / ) ^ 1 ^ 2 * EM 0)
def lipB ( b r) (h h k k) (Mx ) := (uWeight + vWeight) * ((1 - r / ) ^ 1 ^ 2 * EM 0)
  + cWeight * ((1 - r / ) ^ 1 ^ 2 * EM 1)
theorem abs_canonA_ampCoeff_sub_le ... : |A_ (x,y) - A_ (x',y')| lipA * (Dy + 2 My Dx)
theorem abs_canonB_ampCoeff_sub_le ... : |B_ (x,y) - B_ (x',y')| lipB * (Dy + 2 My Dx)
```

Here $D_y = \|y - y'\|_\rho$, $D_x = \|x - x'\|_\rho$ and $\text{EM } \alpha \ell = \int_0^\infty s^{\alpha-1} (1 + |\log s|)^\ell (1 + s) e^{2\beta M_x s} e^{-\beta s^2} ds$.

2 Mathematics

Unit 143 gives, for $s > 0$, $|c_{ij}(s) - c'_{ij}(s)|\rho^{i+j} \leq e^{2\beta s M_x} (D_y + 2\beta s M_y D_x) \leq (D_y + 2\beta M_y D_x)(1 + s)e^{2\beta M_x s}$, a Gaussian envelope of the type `gaussEnv` $\beta (2M_x)$ 1. Its weighted tail moment is therefore

at most $\rho^{-(i+j)}(D_y + 2\beta M_y D_x)\text{EM}$, and summing against r^{i+j} gives the weighted moment norm bound with the geometric factor $\sum_{ij}(r/\rho)^{i+j} = (1 - r/\rho)^{-2}$. The canonical coefficient differences are then bounded by unit 141's difference form, which needs a common envelope for both data sets: $\|y\|e^{\beta s x_{00}}e^{\beta s \|\bar{x}\|} \leq M_y e^{2\beta M_x s}$ on $s \geq 0$ (`ampEnv_le_common`). Astra #11 points out that a norm-level route (integrating $\|c_s - c'_s\|_\rho$ directly against the moment weight, with a sum-integral interchange) would avoid the factor $(1 - r/\rho)^{-2}$; since unit 140's weights already require $r < \rho$, we kept the coordinatewise route, which is what the bounded-linear functionals of unit 140 consume.

3 Paper-facing meaning

The paper's proof of `thm:strataempiricaexpansion` needs $\xi \mapsto C_{\mu,m}(\xi)$ continuous on $C^\omega([0, b]^d)$ and cites `prop:convergence`, which proves only the continuous mapping theorem. For $d = 2$ the present unit supplies the chart-level continuity: the canonical coefficients are locally Lipschitz in the ρ -weighted ℓ^1 norm of the Taylor data (x, y) of (ξ, η) , with constants depending only on $(\beta, b, \rho, r, h, k, \alpha, M_x)$. The bridge to the paper's C^ω topology (Cauchy estimates $\|x\|_\rho \leq (1 - \rho/R)^{-2} \sup_{\bar{D}_R^2} |\xi|$) is a compatibility remark, not formalised. Next: the weighted coefficient space as a rescaled ℓ^1 (`Mathlib lp`), the continuous mapping transfer of convergence in distribution, and the normalised-remainder statements.

4 Lean notes

- `positivity` cannot see $0 \leq M_y$ for a bare variable; derive it from `wnorm_nonneg` and the bound and pass `mul_nonneg` explicitly.
- The two-slot inequality $D_y + 2M_y \beta s D_x \leq (D_y + 2\beta M_y D_x)(1 + s)$ is `nlinarith` with the two product nonnegativity facts supplied.
- $\sum_{ij} \theta^{i+j} = (1 - \theta)^{-2}$ is `tsum_mul_tsum_of_summable_norm` (in `Mathlib.Analysis.Normed.Ring.InfiniteSum`) plus `tsum_geometric_of_lt_one`.
- `abs_canonA_sub_le` accepts the envelope `ampEnv` via `ampCoeff_abs_le` by definitional unfolding, exactly as in unit 119.